

Immune Gene Panel for Infection Response Prediction ¹				
Nineteen genes quantified by high-throughput qPCR at day 8 post-infection ¹				
Gene	Full Name	Primary Function	Category	Response Type
Cell-mediated Immunity				
IRGM1	Immunity-related GTPase M	Autophagy, parasite resistance	Cell-autonomous	Cell-autonomous
MUC2	Mucin 2	Intestinal barrier, pathogen exclusion	Barrier	Barrier
NCR1	Natural cytotoxicity receptor 1	NK cell activation receptor	Cytotoxic	Cytotoxic
PRF1	Perforin 1	Cytolytic granule protein	Cytotoxic	Cytotoxic
Innate Immunity				
TICAM1	TIR domain-containing adapter molecule 1	TLR adaptor, viral response	Innate signaling	Innate
MPO	Myeloperoxidase	Oxidative burst, neutrophil marker	Innate effector	Innate
TLR2	Toll-like receptor 2	Bacterial recognition	Pattern recognition	Innate
TLR4	Toll-like receptor 4	LPS sensor, innate immunity	Pattern recognition	Innate
Regulatory/Anti-inflammatory				
IDO1	Indoleamine 2,3-dioxygenase 1	Tryptophan depletion, immunosuppression	Metabolic enzyme	Regulatory
SOCS1	Suppressor of cytokine signaling 1	Negative feedback, prevents tissue damage	Regulatory	Regulatory
IL1RN	Interleukin 1 receptor antagonist	IL-1 antagonist, limits inflammation	Regulatory	Regulatory
IL-10	Interleukin 10	Anti-inflammatory, regulatory	Regulatory	Regulatory
Source: Webster et al. (2025) <i>in review</i>				

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Gene	Full Name	Primary Function	Category	Response Type
ARG1	Arginase 1	M2 macrophage marker, tissue repair	Metabolic enzyme	Th2/Regulatory
Th2/Anti-helminth Response				
RETNLB	Resistin-like beta	Th2 effector, goblet cell hyperplasia	Th2 response	Th2
IL-13	Interleukin 13	Th2 cytokine, mucus production	Th2 response	Th2
Th1/Pro-inflammatory Response				
CXCL9	C-X-C motif chemokine ligand 9	T cell recruitment, IFN- γ -inducible	Chemokine	Th1
TNF	Tumor necrosis factor	Pro-inflammatory, induces cachexia	Cytokine	Inflammatory
IFNγ	Interferon gamma	Th1 master regulator, parasite control	Cytokine	Th1
IL-6	Interleukin 6	Acute phase response, inflammation	Cytokine	Inflammatory

¹ Genes selected based on established roles in parasite immunity. Expression normalized using quantile normalization with GAPDH and PPIB as housekeeping genes.

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